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REVIEW

A survey of feature matching methods

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Abstract
Feature matching plays a crucial role in computer vision, with applications in visual localization, simultaneous localization and mapping (SLAM), image stitching, and more. It establishes correspondences between sets of feature points from multiple images, enabling various tasks. Over the years, feature matching has witnessed significant development, with an increasing number of methods being applied. However, different methods exhibit different degrees of applicability in different scenarios and requirements due to their different rationales. To cope with these issues, a comprehensive analysis and comparison of matching methods are essential. Existing reviews often lack coverage of deep learning models and focus more on feature detection and description, neglecting the matching process. This survey investigates feature detection, description, and matching techniques within the feature-based image-matching pipeline. Representative methods, their mechanisms, and application scenarios are also briefly introduced. In addition, comprehensive evaluations of classical and state-of-the-art methods are conducted through extensive experiments on representative datasets. Particularly, matching-based applications are compared to fully demonstrate the advantages of the methods. Lastly, this survey highlights current problems and development directions in matching methods, serving as a reference for researchers in the field.

1 | INTRODUCTION

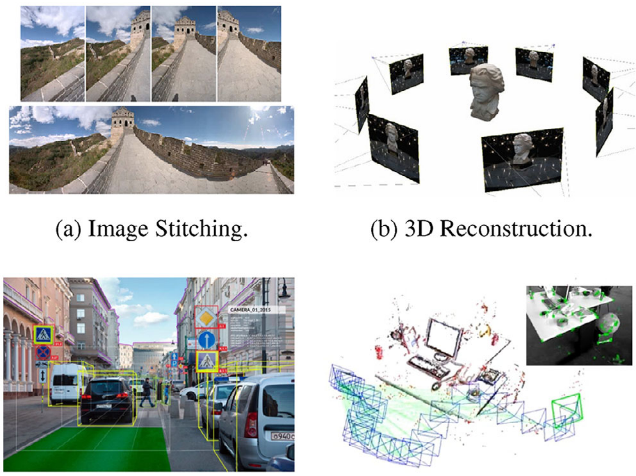
Image matching plays a critical role in computer vision tasks as it seeks to discover one or more transformations within the transformation space. These transformations ensure that multiple images of the same scene, captured at different times, with different sensors, or from varying viewpoints, maintain consistent spatial correspondence. Furthermore, the effectiveness of image matching depends on the presence of shared or similar objects, scenes, or semantics in the images being compared. Figure 1 illustrates the correspondence between the feature points of the rigid (mountain) and non-rigid (cat) objects.

The current image matching has a wide range of application scenarios, as shown in Figure 2, including image retrieval [1, 2], 3D reconstruction [3, 4], simultaneous localization and mapping (SLAM) [5], video stabilization [6, 7], etc. Taking 3D reconstruction as an example, in the field of service robotics research, the aim is to enable robots to autonomously perform movements, avoid obstacles, recognise objects and interact with behaviours in the environment. The basic requirement for solving this problem is for the robot to be able to reconstruct the environment in three dimensions, that is, to determine the distance between the robot and its environmental object points [8]. To accomplish these tasks, identifying and matching pixel points with the same 3D position in an image, that is, image matching, is an important preliminary step, and the robustness and real-time performance of the matching algorithms also affect the subsequent execution of these tasks; thus robust and real-time image matching algorithms are crucial. To date, image matching remains a challenging problem as it requires consideration of multiple factors, including various variables such as point of view, illumination, occlusion and camera attributes.

Depending on what is being matched, current image-matching algorithms can be divided into two categories: region-based methods [9] and feature-based methods [10]. The region-based methods use a direct comparison strategy, which

(a) Rigid object match. (b) Non-rigid object matching.

FIGURE 1 Image matching relationships.



(a) Image Stitching. (b) 3D Reconstruction.

(c) Target Detection. (d) SLAM.

FIGURE 2 Image matching application scenarios. SLAM, simultaneous localization and mapping

directly uses the pixel similarity of the original images in a sliding window of a predetermined size to determine the correspondence between the images. Feature-based methods offer more flexibility and stability and are commonly used in image matching. This approach consists of detecting unique image structures, such as corner and edge points, and then characterising them using a designed mathematical model. Descriptors are then generated for these feature points, thus converting the correspondence problem into finding nearest neighbours in the high-dimensional feature space of the descriptors. It is based on the premise of extracting significant image region features. First the initial matching set construction is carried out, and then the mismatch rejection work is completed. In addition to this, end-to-end matching methods are also an important part of feature matching. However, feature matching also encounters challenges in defining and extracting a high percentage of features that correspond to the same position in 3D space in the real world. Matching N feature points to N feature points detected in another image results in a combinatorial explosion of possible matchings ($N!$), especially when dealing with thousands of features extracted from high-resolution images that often contain outliers and noise. Handling these challenging scenarios has shown promising results with existing methods, albeit at the cost of increased computational time. Consequently, selecting an appropriate method based on different matching scenarios becomes crucial.

In the field of feature matching, early literature reviews [11, 12] mainly focused on traditional manual matching methods. With the emergence of deep learning, it has gradually transformed into a mainstream method for studying feature matching. Unfortunately, there is a lack of comprehensive literature documenting the historical development and application scenarios of neural networks in this field. Currently, existing reviews tend to be limited in scope, focusing either on the examination of feature detectors [13] or the analysis of models specifically tailored to particular application scenarios, such as multimodal image matching [14], synthetic aperture radar (SAR), and optical image matching [15], and 3D augmented reality [16]. Furthermore, while some of the literature [17] attempts to address the complete process of feature matching and incorporates deep learning aspects, it often becomes overly complex and contains too many irrelevant feature matching aspects. Additionally, the experimental sections of these studies are not without flaws, as they lack comprehensive evaluation algorithms and tend to favour comparisons of traditional methods, thus neglecting to evaluate the effectiveness of the proposed methods in downstream computer vision tasks.

The main contributions of this paper compared to the existing literature review are as follows:

(1) We explore the problem of feature-based image matching and fill a gap in the current review of learning-based matching methods. We provide operational strategies for feature matching and conduct an extensive study of both traditional and deep learning based methods. Focusing on feature extraction, mismatch rejection and end-to-end matching methods, we provide a comprehensive overview of the field, as shown in Figure 3.
(2) We present a comparative analysis of classical methods and state-of-the-art methods, combining traditional and deep learning techniques. The performance evaluation is performed on three well-established datasets. By attributing the experimental results to the internal principles and characteristics of each method, we aim to reveal the strengths and limitations of these approaches.
(3) In addition to traditional evaluation metrics, we introduce a comparative analysis of methods for two specific application directions: pose estimation and visual localization. The aim is to investigate the applicability of deep learning methods in these application areas and to evaluate the potential application prospects of various neural networks in different downstream tasks.
(4) Based on the current mainstream research progress, we comprehensively summarise the challenges that still exist in the field of feature matching. Furthermore, we discuss in detail the current challenges of feature matching and explore its potential future directions.

Section 2 of this paper describes the various types of methods used in the feature extraction phase, and their development history and current status. Section 3 focuses on the methods used in the feature matching phase, including initial match set construction, mismatch rejection, and end-to-end feature matching.

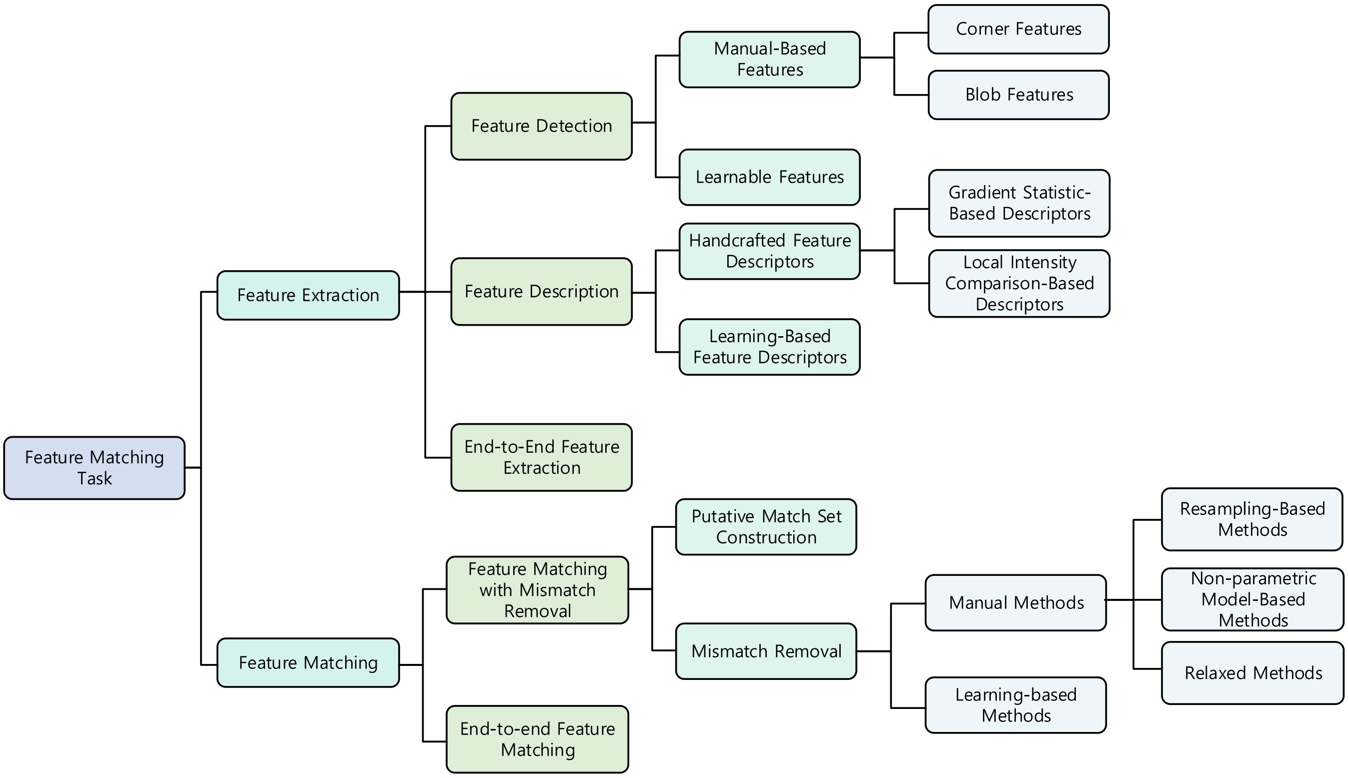


FIGURE 3 Classification of feature matching methods.

In Section 4, the mainstream datasets and evaluation criteria are summarized, followed by a detailed comparison of the main methods mentioned in the paper. Section 5 explores the current challenges and future research directions in feature matching. Finally, Section 6 provides a comprehensive summary of the whole paper.

2 | FEATURE EXTRACTION

Feature extraction is the key stage of the feature matching task. To match a pair of images, the first step is to discover the features in the image using a feature detector. In traditional methods, features can be identified based on changes in pixel intensities in the image. In contrast, deep learning-based methods use convolutional neural networks to generate a feature map that extracts deep or shallow features depending on the particular convolutional layers employed. Once the features are extracted, descriptors are computed based on the coordinates of the features and their neighbourhood relationships. The descriptors are usually in the form of vectors that encapsulate information about the image features related to the key points. Different descriptors have their own strengths and limitations and are therefore suitable for different application scenarios. Additionally, there are end-to-end feature extraction methods that output a set of feature representations directly after inputting the images to be matched. These representations can then be used in the matching task.

2.1 | Feature detection

Initially, image feature extraction relied on manual annotation by human experts, which was costly and inefficient. With the development of computer vision technology, various feature detection methods have been developed based on factors such as the local intensity variations around feature points. These methods include traditional methods and learning-based methods. In traditional methods, feature detection is subdivided into corner-based detection methods and blob-based detection methods based on the specific regions of interest considered for feature detection.

2.1.1 | Manual-based feature detection

(1) Corner features: The earliest method for feature extraction focussed on corner detection, which primarily detects intersections and boundaries. One such method is the greyscale gradient-based extraction method proposed by Moravec [18]. It calculates the second-order gradients in the horizontal, vertical, and two diagonal directions within a moving window and compares them with a threshold. If the minimum value is below the threshold, it is considered an "interest point." Gradient-based detectors tend to rely on the first-order information in the image to locate features.

The original detection method, the Moravec detector, can identify local optima in the entire image, where these

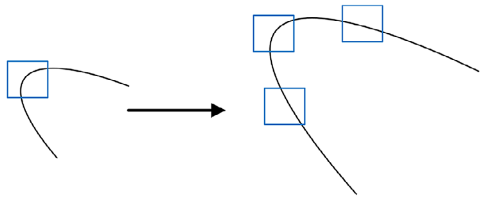


FIGURE 4 Corner points at different scales.

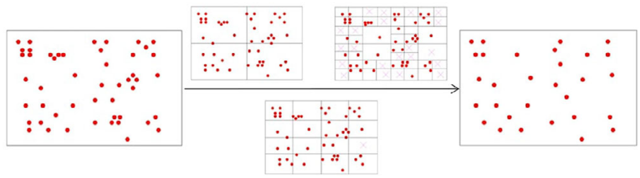


FIGURE 5 Quadtree homogenization for feature extraction.

points exhibit the least similarity to their surrounding environment. However, this detector is not robust when the image undergoes rotation or changes in orientation, displaying instability in such cases. Therefore, building upon the gradient-based approach, Harris et al. [19] introduced Harris corners, with the primary goal of achieving both translational and rotational invariance. They defined corner points based on the rate of intensity changes in two orthogonal directions and further weighted them using a Gaussian function. When the eigenvalues of the intensity matrix exceed a specified threshold, the point is considered a corner. This method overcomes the limitation of Moravec's approach, which only detects corners at multiples of 45°. However, for image matching tasks involving complex variations, the aforementioned model, which only achieves these two invariances, lacks robustness. As shown in Figure 4, corner features at different scales demonstrate the importance of scale invariance. When the image scale changes, a detection window of the same size sliding over a high-resolution image (right) transforms the corner features into line features. This exemplifies the significance of scale invariance.

To address the issue of scale invariance and meet the real-time requirements of most tasks, as well as to fullfill engineering needs, detectors based on pixel intensity have been proposed. The simplest extraction method can be described as follows: comparing the value of a pixel with the values of its surrounding pixels; if the difference is below a certain threshold, the pixel can be considered a non-feature point. Smith et al. [20] introduced the small univalve segment assimilating nucleus (SUSAN) algorithm, which computes the region called the univalve segment assimilating nucleus (USAN) within a circular template that encompasses pixels with similar intensities to the core pixel. By subtracting the USAN from a geometric threshold, corner points are obtained. This method does not rely on gradient calculations, resulting in faster computation speed.

However, the simplistic measurement approach of SUSAN tends to produce a large number of false corner points. From an intensity-based perspective, the FAST corner detection method [21] constructs a discrete circle that passes through 16 pixels and directly compares the differences between the central pixel and the surrounding pixels. If the difference between the pixel values of 12 consecutive points on the circle and the pixel value at the circle centre exceeds a threshold, the circle centre is considered a corner feature point. Although the features from accelerated segment test (FAST) method does not achieve subpixel-level localization accuracy for feature points, in specific application scenarios, speed often takes precedence over high localization accuracy. Building upon the FAST concept, the ORB-SLAM1 detector [22] enhances the specificity of feature descriptors by comparing the intensities of two concentric circles with certain geometric constraints. In addition to employing non-maximum suppression to remove adjacent feature points, ORB-SLAM2 [23] utilizes a quadtree structure to evenly distribute the extraction of features. This general approach leads to a more balanced distribution of feature points, as illustrated in Figure 5.

Currently, available corner detection methods based on local structure tensors fail to adequately describe the intensity variation differences between edges and corners, resulting in erroneous corner detection. Zhang [24] proposed the use of the anisotropic Gaussian directional derivative filter (FOAGDD) and extracted intensity variations from the input image in multiple scales. This technique accurately characterizes edge and corner features in the continuous domain. Building upon the FOAGDD method, Wang [25] smooths the input image using multi-scale, multi-directional shear filters and constructs multi-scale, multi-directional structure tensors for corner detection. This approach exhibits superior detection accuracy, localization precision, and robustness to affine transformations, illumination changes, noise, viewpoint variations, and other factors compared to existing corner and interest point detectors. Although these detection methods can extract feature points robustly under complex transformations, they still suffer from issues such as uneven distribution of feature points and limitations imposed by image scale variations, as corners often exist along object or image boundaries.

(2) Blob features: In contrast to corner detection, blob features refer to closed regions with a specific size, where the pixels within the region are similar to each other but differ from the pixels outside the region. Building on the concept of spatial scale theory [26], feature point extraction algorithms with scale invariance have been developed. The most classical handcrafted feature extraction algorithm, scale invariant feature transform (SIFT) [27], is based on the local appearance interest points of objects. It exhibits scale and rotation invariance and adapts to a certain degree of affine transformations.

SIFT utilizes a DoG image pyramid to create a scale space, which simulates the appearance variations of extracted feature

FIGURE 6 Proportional space based on Gaussian kernel.

points at different image scales through the use of Gaussian functions and downsampling. Figure 6 provides a simple example of applying the Gaussian kernel to generate a scale space. The image on the right has a larger scale and more detailed features compared to the image on the left. By combining a large-scale factor with downsampling, the coarse features of the left image can be approximated. Next, SIFT employs second-order derivatives to identify corner feature points that possess invariance to orientation and image size. Building upon SIFT, more robust detectors have been proposed, such as the affine-SIFT (ASIFT) [28], which has affine invariance, and the scale-invariant feature error (SIFER) [29], which has scale invariance. However, these detectors share the same issue of high computational complexity as SIFT, which is inherent to the Gaussian feature descriptor method itself.

Due to its simplicity and fast computation, SIFT is still commonly used in various feature matching tasks, particularly in scenarios where high accuracy is not required. On the other hand, ORB-SLAM, as an extension of FAST, can extract more abundant and evenly distributed feature points by incorporating pixel intensity information, resulting in higher robustness that is unaffected by different environmental conditions. However, traditional feature descriptors also have limited expressive power, which poses challenges in scenarios involving significant deformations in images used for object localization and tracking, for instance. Furthermore, relying solely on explicit mathematical transformations to analyse greyscale differences between images for feature extraction becomes impractical under varying illumination conditions.

Meanwhile, in contrast to manual crafting methods that rely on domain expertise to detect and describe robust feature points, approaches based on machine learning or deep learning have emerged as research hotspots. The advancement of deep learning theory and high-performance computing has made deep architecture-based feature point extraction a popular research direction. These methods do not rely on explicit handcrafted features but leverage the knowledge of deep learning algorithms, which can overcome the limitations of traditional approaches.

2.1.2 | Learning-based feature detection

Currently, learning-based feature detection methods [30, 31] conceptually map an image onto a score map obtained through machine learning training. The value (score) of each pixel can be interpreted as the probability of it being a salient feature. In particular, deep learning models accumulate image features from shallow to deep layers, progressively merging and refining low-level features, ultimately yielding deep and abstract features at the semantic level.

Lenc proposed a generic framework for learning local covariant feature detectors called DetNet [32], which addresses the problem of extracting viewpoint-invariant features from images. It treats the feature point detection problem as a regression problem and incorporates deep neural networks to assist in regression, enabling successful learning of corner and orientation detectors. To handle situations with severe changes in weather and lighting conditions, Verdie [31] trained segmented linear regressors to predict score maps, and by applying simple non-maximum suppression, they identified good key points with strong generalization capabilities. Detone introduced MagicPoint [33], a convolutional network that generates a heatmap identical to the input image. Keypoints with higher responses in the heatmap are prioritized for detection. Compared to other methods, MagicPoint can successfully detect corners in visual conditions with various types of noise.

To achieve more accurate feature detection, Simo-Serra [34] introduced random sparse connections and hard negative mining between network layers to improve speed and prediction accuracy. Considering deep learning is a black box, and improving accuracy in only the first stage may not enhance the overall task, Bhowmi [35] embedded the detection framework into a complete visual task pipeline and trained the learnable parameters in an end-to-end manner. Key.Net [36] combines handcrafted and learned convolutional neural networks (CNN) filters, where handcrafted filters provide anchor structures for the learned filters to localize, score, and rank features. Key points are then detected at different scales using the network to enhance the detection score. However, Key.Net performs poorly in matching rotated images in practical applications. To address this limitation, Lee [37] proposed a self-supervised orientation key point detection method using rotation-equivariant CNN. A separately pooled rotation-equivariant representation is used to generate robust features for orientation key point detection, allowing the detector to handle various non-rigid transformations more effectively. Attention modules are commonly utilized for feature extraction in various applications. In the study by [38], two attention modules, namely arbitrary-oriented spatial pooling (ASP) and scalable frequency pooling (SFP), are proposed to enhance feature extraction and improve classification accuracy. The work presented by [39] introduces context-aware deformable transformers for end-to-end chest abnormality detection (CDT-CAD), which involves the construction of an iterative context-aware feature extractor. This feature extractor not only expands the receptive fields using dilated context encoding blocks to encode multi-scale context information but also captures distinctive and adaptable patterns of feature variation in the wavelet frequency domain through the use of frequency pooling blocks.

Deep learning has brought new advancements to feature detection, as learning-based detectors can identify key points that may go unnoticed by traditional methods. Moreover, these detectors can be jointly optimized with downstream matching

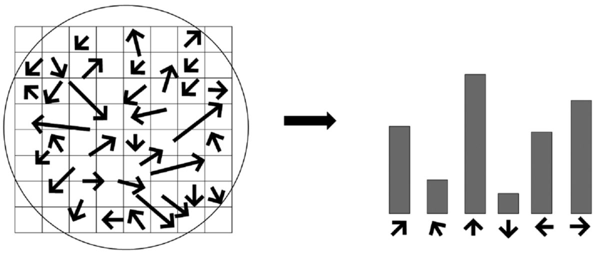


FIGURE 7 Acale invariant feature transform main direction.

applications (e.g., defect detection [40]) to improve matching performance. However, these detection methods require large-scale training sets, and if the images in the matching task significantly differ from the training set, inaccuracies may arise. While deep learning exhibits prominent advantages, its drawbacks, such as the need for better hardware and high-quality training datasets, are also evident. Currently, it can only be applied to matching tasks with high computational resources, and further efforts are required to overcome this limitation and enhance the usability of these methods.

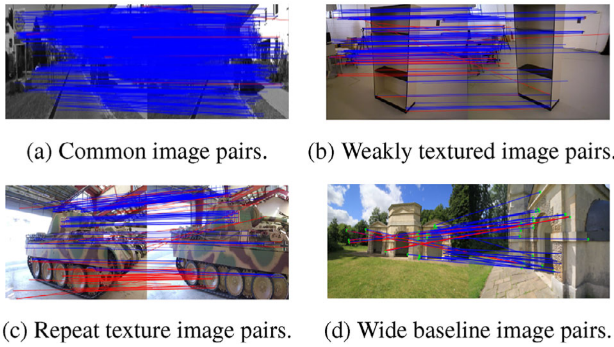
2.2 | Feature description

After detecting key points from the original image, the feature point descriptor transforms the detected key points into coordinates in the feature descriptor space. In this feature descriptor space, ideally, different feature points are far apart, while similar features are closer together. Typically, feature point descriptors extract low-level information from local image regions based on a local window around the feature points. According to the type of extracted information, descriptor methods can be divided into descriptors based on gradient statistics and descriptors based on local intensity comparison. The collected low-level information is then aggregated to obtain descriptors. In addition to traditional descriptor acquisition methods, learning-based descriptors have recently become more and more popular.

(1) Gradient statistic-based descriptors: Gradient statistic-based methods are commonly used to compute floating-point descriptors, and the most classical handcrafted descriptor, SIFT [27], is an example of a floating-point descriptor. Due to its scale and rotation invariance, as well as its robustness to certain affine deformations, SIFT is still widely used today. SIFT computes the gradient direction and magnitude for all pixels within a designated region, and the dominant orientation of the feature point is determined by the peak of the gradient orientation histogram, as illustrated in Figure 7. Then, a square window of size 16×16 centred at the feature point is divided into 4×4 smaller squares. Within each small square, the gradients of pixels are further calculated in eight different orientations, resulting in a 128-dimensional floating-point feature descriptor. Building upon SIFT, PCA-SIFT [41] reduces the dimensionality of the descriptors using principal component analysis, making the dimensionality variable. Speeded up robust features (SURF)[42] employs Haar wavelet responses as the basic transforms and further processes the greyscale values to generate a 64-dimensional descriptor, offering accelerated computation compared to SIFT descriptors. These two descriptors are still widely used in various visual tasks due to their excellent computational performance and robustness.

In standard SIFT descriptors, the L2 norm of the dominant orientation can vary due to differences in image brightness, leading to poor matching performance between images with significant variations in brightness. To address this issue, ROOTSIFT [43] introduces square root normalization to reduce the differences in L2 norm, making the descriptor less sensitive to noise and adaptive to changes in image brightness. These gradient descriptors based on SIFT have the advantages of simplicity and ease of implementation. However, due to the floating-point nature of the resulting descriptor vectors, they occupy a large amount of memory, making it challenging to achieve fast-matching requirements.

(2) Local intensity comparison-based descriptors: Descriptors based on local intensity comparisons are also referred to as binary descriptors, and their core idea lies in the selection rules for comparisons. In the generation of binary descriptors, the greyscale values of pixels within a window are typically compared, and the comparison results are stored as binary values to simplify computation and reduce storage requirements. Due to their relatively simple structure, these descriptors lack uniqueness and are often used for short baseline matching. The most renowned algorithm for binary robust independent elementary features (BRIEF) [44] randomly selects several pairs of points near the feature point within the window and combines the results of greyscale value comparisons from these pairs. Building upon BRIEF, Rublee et al. [22] incorporated orientation estimation capabilities and computed descriptors during a greedy search process, ultimately forming a 256-byte binary feature descriptor that significantly saves computational time for feature matching tasks. To address the challenges of searching on large-scale images, considering the accuracy, efficiency, and memory consumption of descriptors, the vector of locally aggregated descriptors (VLAD) descriptor [45] aggregates local image descriptors into a vector of finite dimensions and subsequently optimizes the vector by reducing its dimensionality while maximizing the preservation of its distinctive features. This approach substantially reduces the time required for retrieval feature processing and applies to large-scale image feature handling. Although the aforementioned traditional handcrafted fixed feature descriptors achieve satisfactory performance in many applications, their robustness and accuracy exhibit significant degradation in certain scenarios, particularly in the presence of common challenges such as motion blur, weak structures, wide baselines, or low texture, as illustrated in Figure 8. To address this issue, researchers have specifically designed descriptors for particular challenging cases. For instance, Tombari et al. [46] developed more robust



(a) Common image pairs. (b) Weakly textured image pairs.

(c) Repeat texture image pairs. (d) Wide baseline image pairs.

FIGURE 8 Comparison of matching results in different situations.

descriptors for identifying textureless image pairs, while Lee et al. [47] proposed a descriptor with blur invariance by combining integral projections in four directions. However, these targeted descriptors only focus on specific challenging conditions and require extensive experimentation to enhance their matching robustness against various challenging factors. Moreover, improving the balance between speed and accuracy in real-world applications becomes increasingly challenging.

2.2.1 | Learning-based feature descriptors

Meanwhile, deep learning approaches have demonstrated remarkable accuracy by replacing many image processing and computer vision techniques, leading researchers to gradually shift their focus towards deep learning-based descriptor generation methods. The earliest instance can be traced back to Osendorfer et al. [48], who proposed training descriptors using Siamese convolutional neural networks. They mapped input image patches into low-dimensional descriptors, where smaller distances between descriptors indicated greater similarity between image patches. However, this learning approach requires the absence of mismatched image pairs in the training set, which contradicts the practical matching tasks that require distinguishing more mismatched pairs. To address this issue, Tian et al. [49] introduced the L2-Net network, that employs the “hardest negative mining” strategy for loss computation. Compared to handcrafted descriptors, learning-based methods do not rely on explicit models and are more amenable to design and improvement through trainable parameters. Zagoruyko et al. [50] designed and tested various network architecture variants to provide references. Building upon VLAD, Sheng proposed the Weighted VLAD (W-VLAD) [51], which incorporates different coefficient weights to account for variations between descriptors and clustering. This allows for better utilization of semantic information and finds application in crowd images. Similarly, the NetVLAD descriptor [1], based on VLAD, employs a generalized VLAD layer with learnable parameters and achieves promising results in visual localization through metric learning strategies.

SuperGlue [52] employs attention-based Graph Neural Networks (GNNs) to find correspondences between feature points in two images. It utilizes self-attention (enhancing the receptive field of feature descriptors) and cross-attention mechanisms (facilitating cross-communication) to mimic human observation during image matching. This approach represents an important milestone in achieving end-to-end deep SLAM. Similarly, inspired by human behaviour of comparing global contextual information in addition to comparing local salient features in matching ambiguous regions, Luo et al. [53] design a unified learning framework to enhance local descriptors using visual context from higher-level image representations and geometric context from the distribution of 2D key points. Motivated by the integration of global contextual information, Sun et al. [54] propose LoFTR (local feature transformer), a detector-free matcher that establishes accurate, semi-dense correspondences in a coarse-to-fine manner through transformer-based transformations. However, these networks have limited image resolution due to memory constraints. To address this, Zhou et al. [55] introduce Patch2Pix, a refinement network that refines matching establishment by regressing pixel-level correspondences from local regions. For the problem of loop closure detection in the visual domain, the trained convolutional autoencoder for loop closure (CALC) network [56] constructs a global feature space to describe the visual appearance and semantic layout of images. It extracts feature descriptors from key point locations within the highest activation regions of the bottom-level convolutional mappings. The loop closure detection system based on these descriptors demonstrates high robustness against false positives.

Descriptor-based approaches that rely on learning are typically trained on limited datasets, and their performance is heavily influenced by the training samples. While they outperform traditional descriptors, they exhibit poor performance in scenarios with blurred or weak structures. End-to-end feature extraction, which combines feature detection and description, can be improved through joint optimization and other techniques.

2.3 | End-to-end feature extraction

Due to the powerful learning and inference capabilities of deep learning, detection and description can be achieved simultaneously. By employing end-to-end deep networks, key point detection, and its descriptors can be learned directly from the entire image, thus better exploiting the relationship between key points and descriptors.

Learned invariant feature transform (LIFT) [57] achieves simultaneous feature detection, orientation estimation, and feature description, with each component implemented and fused based on convolutional neural networks. It undergoes stepwise training to optimize the different objectives of each component, resulting in more discriminative descriptors. In contrast to LIFT, SuperPoint [58] is an end-to-end network trained through self-supervision. It pioneers the creation of a synthetic dataset to train a convolutional network for the joint detection of feature points and their corresponding descriptors in the feedforward process. Moreover, based on the experimental self-supervision information from this work, it is observed that

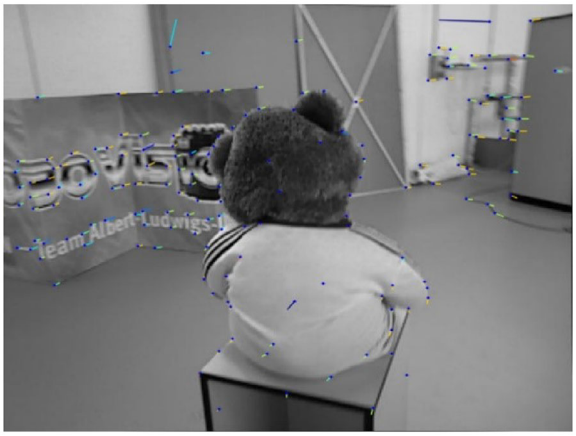


FIGURE 9 Deep learning-based feature point detection (using SuperPoint).

deep learning methods do not necessarily select feature points with "rapid local changes" as designed by handcrafted methods but rather predict repeatable points. This is achieved through the utilization of multiple layers with a large receptive field and pooling operations, allowing for each point in the feature map to have a much larger receptive field than in handcrafted methods, enabling the detection of feature points even in smooth regions of an image, as shown in Figure 9. In the case of complex imaging conditions, the SuperPoint descriptor still exhibits feature extraction instability. D2-Net [59] defers the feature detection stage instead of relying on low-level information early on. It first computes a set of feature maps using CNN and then uses these feature maps to compute descriptors and detect key points. This tight coupling between the detector and descriptor achieves high performance in scenes with variations in lighting conditions, such as day-night transitions, and weak textures. On the other hand, R2D2 [60] takes an opposite approach to generating feature descriptors. It posits that learning descriptors only in regions with high confidence matches contributes to improved performance. R2D2's self-supervised detection and description method simultaneously outputs sparse, repeatable, and reliable key points, thereby avoiding losses incurred by learning in blurry regions.

The aforementioned methods, as dense feature extractors, possess the ability to estimate local shape attributes (scale, orientation, etc.) of feature points. But shape awareness is crucial for geometric invariance. ASLFeat [61], building upon D2-Net, adopts a variational convolutional network to densely estimate and apply local transformations. It leverages the inherent feature hierarchy to recover spatial resolution and low-level details, enabling precise key point localization and enhancing the geometric invariance of descriptors.

Currently, feature extraction techniques are quite sophisticated. However, problems such as noise may occur when using these methods to extract features from images. Furthermore, after obtaining features through extraction, matching N source features with N target features results in $N!$ possible matches. Therefore, establishing a reasonable matching set and rejecting mismatches are crucial for image matching.

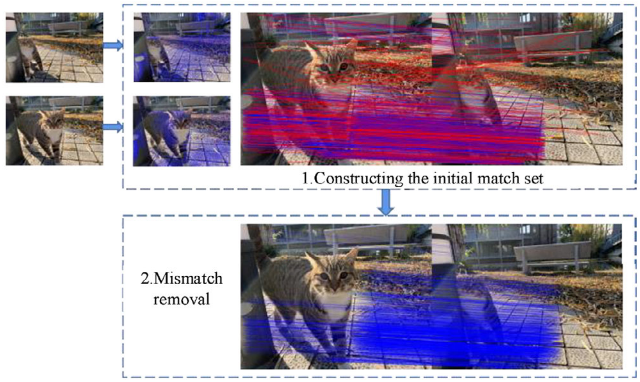


FIGURE 10 Feature point-based image matching methods.

3 | FEATURE MATCHING

After extracting feature points and obtaining feature descriptors, the next steps involve constructing an initial matching set and identifying and removing non-matches from this set. This process follows a two-step strategy common in feature matching. In the first step, feature points are extracted from the image and described, and then the initial matching set is constructed using the generated feature descriptors. In the second step, mismatches are eliminated using geometric consistency constraints. The specific implementation process is shown in Figure 10.

The feature matching step faces three main challenges: (1) Uneven distribution of initial matching pairs across different image pairs due to descriptor limitations. For instance, if the image pairs are captured in indoor scenes, it will result in a sparse distribution of matching pairs. Moreover, if the quality of the image pairs is low, the proportion of incorrect matches in the initial matching set will also be high. (2) Image pairs undergo various transformations and have different requirements for visual tasks. This requires the application of predefined transformation models that are limited to specific scenarios, whereas general models perform poorly in image pairs with complex transformations. (3) Balancing quality and speed is an important consideration in practical applications [62].

To address these challenges, various approaches have been proposed, including traditional methods based on resampling, non-parametric models, and relaxed geometric constraints, as well as mismatch elimination strategies based on deep learning. In addition to the traditional two-step strategy, end-to-end matching methods have shown good performance and have become a new direction for building accurate matching sets.

3.1 | Feature matching with mismatch removal

To match a pair of images, after the feature extraction phase (detecting and extracting local features from the images), the features need to be populated according to the descriptors. Then, using the content of the descriptors, the features in the two images are matched to form an initial set of putative matches, which includes both true matches and mismatches. In the mismatch elimination phase, mismatches need to be eliminated. After obtaining the initial set of matches, the corresponding pairs in the set of putative matches are checked to remove the wrong matches.

3.1.1 | Putative match set construction

The formation of the initial matching set is mainly based on the nearest neighbour (NN) matching strategy [63]. To perform matching, the feature vectors of each image need to be computed, which can be generated using feature descriptors such as SIFT or SURF. These feature vectors are then stored in a feature vector database. For each query feature vector, the nearest neighbour matching algorithm calculates the distances between that vector and all feature vectors in the database, selecting the closest vector as the match. The distance is typically calculated using Euclidean distance or cosine similarity. Although NN is commonly used in tasks such as image matching, object tracking, and 3D reconstruction, it suffers from the problems of local optima and matching errors due to its strong locality.

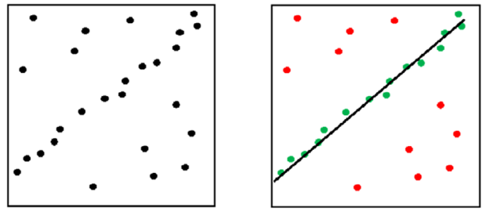
Based on the idea of nearest neighbour matching, Lowe also introduced MNN [63], which is a feature matching method based on the maximum nearest-neighbour distance. For each key point, MNN retains both the nearest neighbour distance and the second nearest neighbour distance. It then computes a ratio by dividing the nearest neighbour distance by the second nearest neighbour distance. The key point is considered a good match if the ratio is below a given threshold. Using this approach, MNN can identify matching key points between two images and use them to calculate the transformation matrix between the images.

Compared to NN, MNN reduces the influence of outliers and produces more robust matching results by using the maximum nearest-neighbour ratio to filter match points. In NN, a threshold needs to be manually selected to filter match points, which is often challenging and requires empirical knowledge or parameter tuning. MNN, on the other hand, does not require manual threshold selection. Instead, it dynamically selects suitable match points based on the ratio calculated from the nearest and second nearest neighbour distances of each key point, resulting in a more stable selection of appropriate match points.

Once the initial matching set has been constructed using the above algorithms, the correspondences within the set can be filtered and evaluated to eliminate false matches and obtain an accurate matching model.

3.1.2 | Mismatch removal

Methods for eliminating mismatches can be broadly classified into manual methods and deep learning-based methods. Manual methods can be further classified into resampling-based methods, non-parametric model-based methods, and relaxed geometric constraint-based methods, each of which relies on different fundamentals.



(a) dataset. (b) RANSAC fitting.

FIGURE 11 RANdom SAmple Consensus (RANSAC).

(1) Manual methods

(a) Resamping-based methods:

A representative method based on the resampling idea is RANSAC (random sample consensus) [64], as shown in Figure 11. RANSAC selects a random subset from the initial match set multiple times and iteratively validates the quality of the model based on the assumption that a pair of images is coupled by a certain parameterized geometric relationship. Assuming the existence of a sample set P, considering the minimum sampling subset S (of size nn) that estimates the geometric relationship, we use this subset to calculate the initial model. For the complement set $SC = P - S$, we substitute it into the data model. If the error is less than the set threshold t, it is considered an inlier (correct match), and all inliers form the inlier set S^*, whose count is calculated. Then, we compare the size of the inlier set between this iteration and the previous iterations, recording the model with a larger number of inliers until K iterations are completed. The size of the sampling subset, n, and the error threshold, t, can be determined empirically. The number of iterations, K, needs to be computed. Firstly, based on experience, a proportion W representing the number of inliers is determined. Then, the probability of correctly selecting n inliers in K iterations is given by

$$P = 1 - (1 - W^n)^K \tag{1}$$

This probability is also the probability of the RANSAC algorithm correctly estimating the model. Finally, the number of iterations, K can be calculated based on the probability P. RANSAC obtains a global geometric consistency constraint between the two images. However, due to its random sampling nature, RANSAC can only obtain local optima instead of global optima. In some cases, such as when the mismatch ratio is high in the sampled subset, random sampling may miss the correct model. Moreover, RANSAC involves a large number of random sampling and model-fitting operations in the dataset, resulting in high computational complexity.

Building on the effective resampling idea and its simple implementation, many researchers have proposed further improvements to enhance the performance of RANSAC. For example, EVSAC [65] presents a probabilistic parametric model that can assign confidence values for each matching correspondence. It combines prior information about estimation confidence to increase the probability of filtering out subsets with true correspondences and achieve a higher matching speed than RANSAC. NAPSAC [66] extracts samples from gradually increasing neighbourhoods and verifies them step by step, reducing the running time. NAPSAC leverages the fact that correct matches are closer to each other compared to mismatches, while GroupSAC [67] assumes that correct matches are similar to each other and groups the feature points based on their similarities. Based on this, they introduce a new binomial mixture model rather than the simple binomial model of RANSAC. It is more efficient to sample data from a smaller number of groups and groups with more tentative correspondences in a new binomial mixture model.

PROSAC [68], similar to the aforementioned methods, ranks the prior matching probabilities for each point and samples the most promising points for matching, ultimately eliminating points with the lowest match likelihood. However, these methods require manual determination of the threshold for excluding outliers, which needs to be determined based on practical considerations and expertise, thereby increasing the usage threshold. Similarly, based on the RANSAC idea, MAGSAC++ [69] does not require the user to provide an exact noise range but instead uses a relaxed upper bound $\sigma_{\max}$ to define a plausible threshold. The algorithm is essentially an iteratively reweighted least squares (IRLS) M-estimator that does not rely on a noise range but employs a novel termination criterion. In traditional matching methods, a matching feature point in one image corresponds only to its nearest neighbour based on Euclidean distance in the other image, which may propagate errors from initial mismatches in subsequent steps, known as the "repeating pattern matching problem" [70]. To overcome this phenomenon, Simon et al. [34] proposed a novel matching workflow that first performs iterative re-matching to allow unmatched feature points to be further considered in subsequent steps. Then, within each iteration, fine matching is performed for matches exhibiting the same transformation, additional features are extrapolated outside the region, and outlier detection is performed using Delaunay triangulation. By repeating the iterations and eliminating the influence of mismatches through detection, this approach effectively addresses the problem. Although such resampling-based methods are widely used and have demonstrated their effectiveness in visual tasks, they still have some drawbacks. For instance, in theory, the computational time of the models grows exponentially with the increase in the error match rate, and the minimum subset sampling strategy is only applicable to parameterized models and cannot handle complex variations such as non-rigid object motion.

(b) Non-parametric model-based methods: Due to the limitations of parametric models generated by resampling methods, non-parametric approaches have emerged to deal with complex image transformations and non-rigid object motion. The distinguishing feature of these methods is that they use different deformation functions to model image transformations and deal with the mismatch problem in different ways.

In early research, Leordeanu and Hebert [71] proposed a relaxed spectral method that divides correspondences into optimal clusters and constructs an adjacency matrix for the elements in each cluster instead of enforcing one-to-one mapping constraints. They then continue to search for correct matches based on the subordination relationship among the optimal clusters in the adjacency matrix, leading to a global constraint model. In this way, the process of searching for correct matches is transformed into the task of finding the cluster C that maximizes the score. Each cluster is composed of correspondences (i, i') between the data and the model. The score computation for the clusters is described as follows:

$$S = \sum_{a,b \in C} M(a, b) = x^T M x \tag{2}$$

The variable x represents an indicator variable for the cluster C, where x is an indicator vector that can represent cluster C. If a candidate assignment $a \in C$ then $x(a) = 1$. $M(a, b)$ describes the extent to which the geometric relationship is preserved when the model features (i', j') and the data features (i, j) correspond. By computing the eigenvector, the optimal solution, namely the optimal cluster, can be easily found.

To accommodate multiple visual modalities, GS [72] introduced spectral constraints based on the L1 norm as a novel approach. However, the results obtained through relaxed matrix matching are often of inferior quality. To address this, Lin et al. [73] incorporated the smoothness of motion between matched pairs into a likelihood estimation function and obtained precise matches by adjusting the affine motion model. This coherence-based separable constraint aims to discover a coherence-based separable constraint from highly noisy matches, producing high-quality matches across wide baselines and effectively adapting to a large number of outliers while eliminating mismatches but allowing a small amount of motion inconsistency. Similarly, Lin [74] proposed a quadruple-filter coherence based decision boundaries (CODE) to achieve good result sets, demonstrating that the motion smoothness constraint serves as a likelihood estimation for every possible motion of each correspondence. However, this coherence-based separable model algorithm has a high time complexity, limiting its application to real-time tasks.

Based on the theory of motion correlation, vector field consensus (VFC) [75] generates vectors from the pixel positions of image correspondences and constructs a vector field based on these vectors. The mapping function is then found by optimizing the smoothness of the vector field. The solution to the mapping function can be obtained through regularization in the reproducing kernel Hilbert space (RKHS), and the specific formula is shown below:

$$\varepsilon(\mathrm{f}) = \min_{f \in H} \left\{ \sum_{n=1}^{N} \left\| y_n - f(x_n) \right\|^2 + \lambda \| f \|_H^2 \right\}$$

The first part represents the experimental error used for data fitting, while the second part serves as a stabilizer that ensures the smoothness of the vector field $f(y_n = f(x_n))$. The regularization constant λ controls the balance between these two parts. By removing outliers from the vector field, VFC can eliminate almost all of the outliers, although it is computationally intensive.

Based on the Gaussian field criterion for remote sensing image matching, Ma et al. propose gaussian field criterion (GFC) [76]. It models the image transformation using both linear and nonlinear functions and employs sparse approximation for non-linear cases, significantly reducing the computational complexity. In the linear case, a general homography is used to model the transformation. In the non-linear case, the non-rigid functions located in a reproducing kernel Hilbert space are considered. It exhibits significant advantages in scenarios involving low overlap, repetitive patterns, or low-quality remote sensing image matching.

Identifying point correspondence function (ICF) [77] introduces the concept of correspondence functions to identify mismatches. It utilizes regression with support vector machines to capture the overall trend of the data. The basic idea of ICF is to construct two functions that connect corresponding points i and i' between images, ensuring a mapping relationship between them through the functions (f, f') as correspondence functions. The iteratively estimate correspondence function (IECF) iterative function generates the correspondence functions, and through iterative updates, it avoids the influence of mismatches on the mapping relationship by checking for mismatches at each iteration. The efficiency of ICF is affected by the proportion of mismatches and the choice of parameters. The parameter selection controls the size of each influencing subset in the ICF iteration process, which affects the number of iterations and the matching efficiency. However, ICF has shown good performance in matching non-rigid object motion images and is still widely used today.

Due to the necessity of computing motion trends over the entire dataset, non-parametric methods suffer from higher computational complexity compared to RANSAC-like approaches. Moreover, due to the unknown motion model, they are susceptible to noise interference during the computation process. For instance, the model may attempt to fit the noise. Therefore, further optimization of the computations and enhancement of the method's robustness are required to address these challenges.

(c) Relaxed methods: Relaxation of geometric constraints, also known as local geometric constraints, has gained popularity as a research direction in comparison to global constraints. Local geometric consistency, or motion smoothness, is currently a more prominent focus. It can be described as follows: given a pair of images captured from the same scene, if a match between two features is a correct correspondence between the images, the neighbourhood features of this match should also exhibit corresponding relationships, as illustrated in Figure 12. Compared to global constraints, local constraints are more relaxed, allowing for adaptation to scenarios involving wide baselines or non-rigid objects with discontinuous motion. Consequently, methods based on local constraints have emerged as a new and popular research trend. Grid-based motion statistics (GMS) [78] argues that matches should exhibit smoothness, where the corresponding features of points within the same neighbourhood as the correct match should also be nearby. According to the smoothness assumption, if a pair of regions contains (n, m) features to be matched, the distribution model of the score S_i for the neighbourhood of the matching x_i can be expressed as follows:

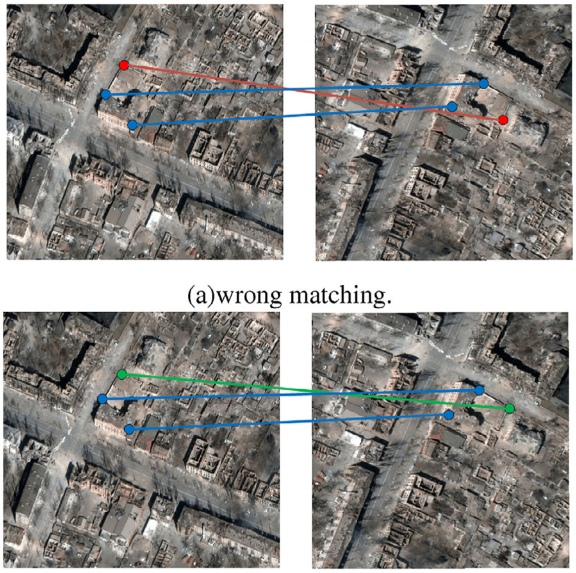
FIGURE 12 Neighbourhood consistency constraints.

$$S_i \sim \begin{cases} B(K_n, p_t), if\ x_i\ is\ true \\ B(K_n, p_t), if\ x_i\ is\ false \end{cases} \tag{3}$$

Where K represents the number of regions that the match i predicts to belong to. In these two binomial distributions, we know that events that are x standard deviations away from the mean are highly unlikely to occur, and this level can be quantified using a partitionability score.

$$p = \frac{m_t - m_f}{s_t + s_f} = \frac{K_{np_t} - K_{np_f}}{\sqrt{Knp_t(1 - p_t)} + \sqrt{Knp_f(1 - p_f)}} \tag{4}$$

m and s represent the mean and variance of the binomial distribution, while p_t and p_f denote the probabilities of inliers and outliers, respectively. The goal of the algorithm is to maximize P to distinguish between inliers and outliers. Additionally, GMS accelerates the process by employing a grid-based approach for region partitioning. It verifies correct matches by examining the neighbouring $n \times n$ grid cells, reducing the time complexity from $O(n^2)$ to $O(n)$ that would otherwise be required for comparing n points. GMS has the advantages of effectively handling a large amount of noise and outliers in images, as well as fast computation



(a)wrong matching.

(b)correct matching.

FIGURE 13 True matches by locality preserving matching.

speed. It has been widely applied in various computer vision applications, such as SLAM and structured light 3D reconstruction. However, due to the grid partitioning nature of GMS, the choice of grid size and shape has a significant impact on the matching results, and it cannot handle scenes with large camera motions.

A more stringent method than GMS is proposed by Ma et al. [79], called locality-preserving matching (LPM), which preserves locality for each correspondence relationship. LPM does not require a specific motion model to estimate global transformations but focuses on preserving neighbourhood structure, making it more versatile than GMS. However, LPM defines the local similarity between feature points by computing the intersection of K-nearest neighbours (K-NN), disregarding the differences between adjacent elements, specifically their specific positional relationships, as illustrated in Figure 13. In this figure, both matches are considered true matches by LPM, but the top match is wrong, while only the bottom match is correct. To address this issue, Jiang proposed TopKRP [80], which converts the potential matching feature points from the feature space to the ranked list space. It measures the topological similarity between two feature points by comparing their ranked lists. The mathematical optimization model can deal with different types of degradation, even when the image pair suffers from large-scale change and rotation. NMRC [81], on the other hand, establishes neighbourhood structures based on manifold representations. It preserves the local neighbourhood structure between two feature points on a low-dimensional manifold and improves matching performance through an iterative filtering strategy based on neighbourhood similarity. Another method, guided neighborhood affine subspace embed (NASE) [82], employs guided neighbourhood affine subspace embedding by preserving the intrinsic manifold of potential true matches. Considering that the corresponding manifold of true matches can be severely affected by outliers, NASE introduces a density-based seed point selection strategy for neighbourhood refinement to enhance robustness. This method achieves linear-logarithmic time complexity for addressing mismatch removal through an iterative optimization framework.

Building upon the concept of spatial clustering, Jiang et al. [83] associated the problem with spatial clustering and utilized DBSCAN as the underlying model. They proposed an adaptive parameter estimation method and an iterative clustering strategy called RFM-SCAN, which achieved nearly linear time complexity. This approach demonstrated excellent performance in applications such as near-duplicate image retrieval. Similar to RFM-SCAN, GLOF [84] also focuses on mismatch suppression. It introduces a guided local outlier factor (GLOF) for robust feature matching, particularly in cases of severe mismatches within multi-scale neighbourhood structures. GLOF effectively mitigates fluctuations in local neighbourhood matching results of different sizes, thus enhancing matching accuracy. The manifestation of multi-granularity lies in the introduction of a series of $K = K = |K_i|K_1, \dots, K_C), i \in [1, C]$ to compute the local correspondence scores (LCS). Each K_i represents a $K_i - \mathit{neighbourhood}$, where M_{k_i} denotes the value of m (the cardinality of matching features) within the neighbourhood. Additionally, p represents the inlier probability within K_i.

$$\begin{aligned} \widehat{LCS}_k(x_i, y_i) &= \frac{1}{C}\sum_{i=1}^{C}\frac{1}{k_i}\left(\left|N/M_{K_i}\right| + \left|\widehat{p}_{k_i}\right|\right) \\ &= \frac{1}{C}\sum_{i=1}^{C}\frac{1}{K_i}\left(K_i - m + \widehat{p}\right) \end{aligned} \tag{5}$$

Based on the LCS, the inliers can be inferred as follows.

$$I^{k}_{x_i, y_i 1} = \begin{cases} 1, if & \widehat{LCS}_K(x_i, y_i) \le \beta, \\ 0, if & \widehat{LCS}_K(x_i, y_i) > \beta. \end{cases} \tag{6}$$

With the advancement of methods based on global and local geometric consistency, AdaLAM [85] emerges as an integrated framework incorporating several classical approaches. It aims to efficiently leverage modern parallel hardware to generate fast and accurate outlier filters. Furthermore, it employs adaptive relaxation assumptions to achieve generalizability across different scenes and extract maximum information from the images. The principle of locality-guided global-preserving optimization (LOGO) [86], contrary to AdaLAM, eliminates the previous methods' limitations on global or local geometric structures. It designs a matching strategy guided by local topological consistency, exploring structural information from local to global scales. Currently, LOGO is primarily applied to tasks such as 3D reconstruction and non-rigid image registration. Local graph structure consensus (LGSC) [87] also utilizes local topological consistency and does not require specific parameters or non-parametric models to explain image transformations. It proposes a multi-scale iterative local graph construction strategy to eliminate mismatches, demonstrating robustness even in the presence of complex non-rigid image transformations. Similarly, leveraging the local geometric structure of features, Liang et al. [88] introduce an affine-invariant feature matching formulation. Under a global transformation model, features are matched by aligning two sets of triplets, significantly improving convergence speed and matching performance, especially in scenarios with high mismatch rates. This method can be used in conjunction with descriptors such as SIFT, harnessing the advantages of the chosen descriptors. Despite the various improvements in recent years, methods based on local geometric constraints still tend to produce local optima rather than global optima in terms of matching solutions. Therefore, a certain level of mismatch may persist when dealing with complex scenes.

Traditional methods have made significant progress in addressing images with complex transformations by evaluating the confidence of matches, estimating motion models, and exploring spatial clustering. Currently, these methods outperform deep learning approaches in terms of computational complexity. However, traditional models still have limitations when dealing with images with unknown geometric transformations or multimodal images, as they struggle to achieve robust adaptability in such scenarios.

(2) Learning-based methods: In recent years, deep learning has gradually entered the field of computer vision and demonstrated excellent performance in applications such as image classification, object detection and tracking, and image segmentation, demonstrating the inherent advantages and robustness of deep learning methods. In feature matching, deep learning has also been widely used for feature detection and descriptor extraction, achieving better results than traditional methods. Neural network-based matchers are still in the developmental stage. Compared with traditional approaches, learning-based mismatch removal methods can effectively utilise contextual information about feature points through strategies such as cross-attention and message passing, thus improving performance. Some PointNet-based methods have shown promising results after further improvement.

PointNet differs from traditional convolutional neural networks (CNNs) as it is a point-based neural network that directly processes point cloud data. Through a series of transformation and extraction networks, PointNet generates a global feature representation. PointCN [89] extends PointNet by adding a CN (context normalization) layer. It formulates the mismatch removal problem as a classifier that distinguishes between inliers (correct matches, i.e. inlier points) and outliers (incorrect matches, i.e. outlier points) by introducing a weighted eight-point algorithm [90] to construct a regression matrix for the input correspondences. To enhance PointCN, N3Net [91] inserts a k-nearest neighbours (KNN) layer into the network, integrating the neighbours into the N3 block, which outperforms other non-local models.

Learning to find good correspondences (LFGC) [89], similarly based on the classifier paradigm, utilizes the weighted formula of the eight-point algorithm to predict correct correspondences and introduces contextual normalization operations. It incorporates global information into each data point while maintaining network invariance, which is simpler and more effective than previous approaches that used global features. Since LFGC learns at the pixel level with a multi-layer perceptron rather than directly operating on images, its structure is simpler and widely applied in the recovery of relative poses between images. However, LFGC requires the input of camera intrinsic parameters, which introduces some barriers to data provision.

ACNE [92] proposes attentional contextual normalization, which trains a perceptron to transform feature maps into attention weights and normalize them, thereby constructing an attentional contextual network. It considers both global and local attention and can identify data points in high-dimensional space that solve the given task, exhibiting excellent performance in pose estimation applications. The recent OANet [93] combines PointCN with geometric deep learning, replacing traditional matching algorithms and RANSAC to handle matched points obtained from feature detection. It captures both global and local features to establish correspondences between image pairs. OANet, based on PointCN, introduces differentiable pooling operations into graph convolutional neural networks to better capture local features. It also adds transformation layers before and after conventional perceptrons to establish connections between points, capture global contextual information, and significantly improve matching accuracy and pose estimation accuracy. However, these methods are based on specific parameter transformation models and have lower generality, making them unable to handle non-rigid images.

To enhance the generality and flexibility of matching, Ma et al. [94] proposed a learnable binary classifier called LMR. The main idea behind LMR is to exploit neighbourhood consistency consensus and eliminate mismatches using a classifier similar to MLP. Since LMR does not rely on geometric parameter models, it can handle different image transformations and can be integrated into matchers that depend on specific transformation models, providing effective initial processing. However, the structure of local neighbourhood consistency is based on certain angles and scale changes and may not handle more complex transformations.

The "local structure visualization-attention" network (LSV-ANet) [13], compared to LMR, is more capable of handling feature matching with multiple variations. This network is based on the steps involved in human visual image matching and employs two attention mechanisms: spatial scale attention (SCA) and spatial structure attention (STA). SCA enriches the neighbourhood information of feature points, while STA is used to form deformation estimation and registration, enhancing the consistency of neighbourhood structures.

The specific mechanisms of SCA and STA are as follows:

$$\mathrm{LSV}_{\mathrm{sca}} = F_{\mathrm{sca}} = \sum_{j=1}^{M} \lambda_j \mathrm{LSV}_{k_j} \tag{7}$$

By utilizing a series of neighbourhood sizes $\{K_j\}_{j=1}^{M}$, a binary image LSV_{K_j} of size 28 × 28 with two channels is generated, where λ_j represents the weight factors to be learned. The application of SCA is highly effective in addressing neighbourhood mismatches by identifying mismatches based on the variation of distances between the neighbourhoods and feature points.

$$\begin{aligned}\left[\mathrm{LSV}_{\mathrm{sca}}^{y_j}\right] &= \sum_{n=1}^{28}\sum_{m=1}^{28}\left[\mathrm{LSV}_{\mathrm{sca}}^{y_i}\right]_n \\ &\max\left(0, 1-\left|\tau_\theta^i - m\right|\right)\max\left(0, 1-\left|\tau_\theta^j - m\right|\right)\end{aligned} \tag{8}$$

The convolutional regression network analyses the corresponding image patches based on $\mathrm{LSV}_{\mathrm{sca}}$ and learns deformation parameters, which are then used for registration by STA. The control points $(\tau_\theta^i, \tau_\theta^j)$ are generated using the predicted deformation parameters and are used to generate the deformation results.

LSV-ANet [95] utilizes multi-LSV descriptors to represent the multi-scale topological structure of feature points and incorporates SCA, STA, and a decision module, allowing for spatial operations on LSV to obtain optimal LSV. Compared to other methods, LSV-ANet requires a smaller number of images to construct the training set and achieves excellent outlier removal performance. However, the LSV descriptors only capture local features for each correspondence segment, neglecting global features. Addressing this limitation, GANet [96] employs an attention mechanism to capture global features. It utilizes a multi-head attention mechanism to capture finer and deeper features, enabling the network to focus more on potential true correspondences while suppressing outliers. To address the memory and computational overhead associated with attention mechanisms, a lightweight attention implementation called Sparse GANet is proposed, which reduces the computational burden while maintaining accuracy.

GNN, unlike PointNet, is a deep learning model used for processing graph-structured data. Distinct from traditional neural network models, GNN treats each node as input and considers the relationships and topological structure between nodes. SuperGlue [52] is an attention-based GNN network and serves as a classic framework for feature matching. It also relies on inputs from feature detection and descriptors, leading to significant improvements in accuracy. SuperGlue constructs a complete graph using key points within and across images and updates node representations through attention-based message passing. However, it has the drawback of requiring both a doubled runtime and space complexity relative to the number of nodes, which becomes evident when dealing with a large number of point matches. To reduce unnecessary point-to-point communication, Shi proposed the ClusterGNN [97] model as an alternative approach for sparse graphs. It introduces a deep learning-based clustering method to establish local graphs for matching and applies the permutation loss based on Sinkhorn networks [98], which is more suitable for feature matching. Each point in the graph interacts only with points in the same local graph, reducing the propagation of redundant information and making message passing more efficient. The network also learns point-level features and structural information (hyperedges), allowing training and testing with different numbers of points in different image pairs and enhancing the robustness of the network.

To further improve performance, SGMNet [99] proposes a seed graph matching network to construct sparse graphs. It initializes the matching by generating a small set of reliable matching seeds and builds a seed graph neural network to propagate messages within and across images and predict matching costs, thereby reducing computational complexity. Inspired by SGMNet, HTMatch [100] adopts a seed GNN architecture but introduces a more efficient message-passing strategy. HTMatch learns both intra-image and inter-image similarities and incorporates a spatial embedding module to embed spatial information from one image into another, enhancing spatial constraints between images and achieving robust matching. The aforementioned GNN-based methods primarily rely on information propagation along edges to form discriminative node representations, and their performance heavily depends on the underlying graph structure, limiting their effectiveness in some challenging scenarios. For complex tasks such as matching images for 3D reconstruction from videos, LifelongGlue [101], a continuous learning neural network, achieves accurate point matching between consecutive images by leveraging knowledge from previous frames. Inspired by SuperGlue, this network introduces continuous self-attention and cross-attention mechanisms to enhance local features, making it well-suited for 3D reconstruction tasks.

Deep learning-based methods for outlier rejection have become a research focus in the field of feature matching. However, these methods suffer from the following issues:

(1) They often rely on the initial set of matches or initial key points, making it challenging to discover additional matches by considering the original matched images.
(2) Some attention-based methods achieve high accuracy but come with high computational and memory costs, especially when dealing with a large number of key points, leading to a significant increase in complexity.

To address these challenges and fully exploit the information in the original images, deep learning has also been applied to end-to-end feature matching, where a pair of images can directly produce a corresponding set of matches.

3.2 | End-to-end feature matching

The end-to-end approach to feature matching consists of using a neural network to handle both the feature extraction and feature matching stages. By outputting the correct matching relationship directly from the input images, this approach differs from traditional two-stage methods. The advantage of the end-to-end approach is that it is able to explore different matching strategies through co-optimisation and train more appropriate feature representations, thus increasing the likelihood of finding a correct match.

Recently, graph convolutional neural networks (GCNs) have demonstrated significant potential in graph matching tasks by integrating graph node embedding, node affinity learning, and matching optimization within an end-to-end model. By approximating the first-order spectral filters, Kipf et al. [102] proposed a simple graph convolutional network (GCN) for graph node representation and semi-supervised learning. To apply this model to graph matching problems, Nowak [103] introduced a Siamese GNN encoder that generates normalized node embeddings for each pair of graphs to be matched and then predicts matching solutions by minimizing the cosine distance between the generated embeddings. Based on this model, Wang [104] proposed GCNs, which optimize graph node embeddings and perform matching predictions within the same network. Once optimal embeddings are achieved, the graph-matching prediction can be approximated as a linear assignment problem, which can be solved using Sinkhorn operations [105]. Existing GCN-based matching methods employ several smooth graph convolutional layers to generate graph node embeddings. However, extensive smoothing operations may dilute the discriminative information of graph nodes. To overcome these issues, GLMNet by Bo Jiang [106] integrated graph learning into graph matching, adaptively learning the best pair of graphs to better serve the matching task. It further introduced a Laplacian sharpening convolutional module to generate more discriminative node embeddings for graph matching. Additionally, the introduced constraint regularization loss encodes the one-to-one matching constraint, aiding in guiding more accurate matching predictions. GLMNet can be used not only for matching between two images but also for multi-graph matching, making it suitable for tasks such as 3D reconstruction.

Since methods like GLMNet rely on messages passing along edges to form discriminative node representations, their performance heavily relies on the graph structure. Furthermore, the heuristic generation of graph patterns may introduce unreliable relationships, leading to performance degradation. To enhance the robustness of different images, Fey et al. [104] proposed an end-to-end differentiable deep network flow for learning the similarity of graph matching. They employed a deep graph embedding model to parameterize the similarity function within and across graphs, effectively capturing higher-order structures. Since this network is category-agnostic, it exhibits cross-category generality and is more applicable to real-world scenarios. In contrast to category-agnostic networks, Graph Learning and Matching (GLAM) proposed by Liu [107] performs classification during training, combining semantic information to achieve higher accuracy. GLAM utilizes a pure attention-based framework for graph learning and matching. It discovers relationships between features through self-attention mechanisms, further updating feature representations within the learned structure. Cross-attention calculates the relationship between two feature sets to perform feature reconstruction and directly derive the matching solution. GLAM's innovation lies in proving that replacing manually designed graph structures with learned graph structures can significantly enhance matching performance.

NCNet [108] emulates the standard steps of feature extraction, matching, simultaneous inlier detection, and model parameter estimation. It enables end-to-end training, and the parameters used during training can be learned from synthetic images, eliminating the need for manual annotation. NCNet employs a twin structure of convolutions to propagate the input images, extracting feature maps that resemble dense local descriptors. The feature maps (descriptors) form initial correspondences between images, and a regression network robustly outputs the parameters of the geometric model, such as a 6-dimensional vector for affine transformation. It uses correlation layers to perform matching within the network and improves the matching scores further through neighbourhood consensus scoring. However, due to memory constraints, NCNet computes scores on feature maps with reduced resolution by a factor of 16, which leads to limited accuracy in camera pose estimation applications. On the other hand, SparseNCNet [109] represents correlation tensors by storing the top ten similarity scores and employs sparse convolutions instead of dense 4D convolutions, achieving matching on feature maps with only four times reduced resolution. By utilizing sparsity in processing, this network allows higher-resolution images as input, enhancing local accuracy.

Subsequently, DualRC-Net [110] extracts feature maps at coarse and fine resolutions. The coarse feature maps are used to generate a complete but rough 4D correlation tensor, which is then refined through a learnable neighbourhood consistency module. The fine-resolution feature maps are utilized to obtain the final dense correspondences guided by the improved coarse 4D correlation tensor. The selected coarse-resolution matching scores allow the fine-resolution features to focus only on a limited number of potential matches with high confidence. In this way, DualRC-Net significantly enhances matching reliability and localization accuracy while avoiding the computationally expensive 4D convolutions on the fine-resolution feature maps, resulting in superior performance compared to SparseNCNet. Both NCNet and SparseNCNet employ weakly supervised losses, where all non-matching pairs are assigned low scores and matching pairs are assigned high scores, which do not effectively distinguish between good and bad matches and are not suitable for pixel-level correspondence localization.

To address this issue, Patch2Pix [55] provides an effective solution. It utilizes a correspondence network in the first stage to obtain block-level correspondences and subsequently refines the correspondences in a refinement network to achieve pixel-level matching accuracy. Finally, a weak epipolar line-supervised loss is introduced to train the network to detect geometrically consistent correspondences, resulting in improved relative pose estimation.

The level of supervision greatly influences the matching results. Learning to guide local feature matches (LGLFM) [111], based on the NCNet architecture, serves as a guided feature matching method. The core idea is to use a deep learning model to predict coarse image correspondences. The study compares weak image-level supervision, weak epipolar line supervision, and point supervision and finds that weak epipolar line supervision outperforms point supervision as an improvement over weak image-level supervision. Ni [112] proposes a self-supervised block transportation matching method called PATS (patch area transportation with subdivision), which is similar to Patch2Pix. It initially divides the original image pairs into blocks of the same size and gradually refines them into smaller blocks of the same scale. PATS's patch area transportation enables self-supervised learning of scale differences, leading to accurate scale estimation. It improves matching accuracy and coverage, thus assisting with downstream tasks such as relative pose estimation, visual localization, and optical flow estimation.

Sparse-to-dense matching, as an asymmetric matching approach, allows for a comprehensive search for correspondences on the target image by performing cross-operations between sparse local descriptors and dense feature maps. S2DNet [113] introduces S2DHM, which stands for Sparse-to-Dense Hypercolumn Matching, capable of providing pixel-level correspondences. It formulates the sparse-to-dense matching problem as a supervised classification task, leading to more robust results in images with illumination variations. Compared to other sparse-to-sparse matching methods, S2DNet can discover a greater number of key points.

Similar to the idea of combining sparse and dense matching, correspondence transformer (COTR) [114] enables finding correspondences between a given query point in one image and its counterpart in another image. This approach allows for sparse correspondence retrieval by querying key points as well as dense mapping by querying all points in the reference image. Moreover, COTR leverages transformer to successfully align images under conditions where optical flow exhibits non-smooth variations at object boundaries.

Whether utilizing neural networks only in the matching stage or directly for end-to-end matching, both approaches exhibit higher generality compared to traditional methods. Table 1 compares the traditional and deep learning-based approaches in terms of mechanisms, advantages and disadvantages, and applicable scenarios.

However, deep learning-based matching methods can have significant differences in performance metrics due to the different feature detection and description algorithms used, resulting in relatively weak robustness. Given the current neural network matching methods, descriptors that are more suitable for achieving end-to-end feature matching can be designed to give full play to their superiority. Furthermore, with the development of deep learning, novel feature detection and description methods based on neural networks are constantly emerging. However, due to the high dependence of deep learning on training datasets, robust and efficient algorithmic models are needed to effectively handle different detection and descriptor scenarios.

4 | EXPERIMENTS

With the improvement of traditional algorithms and the development of the deep learning field, more and more image feature matching methods have been proposed. Due to the different basic principles or implementation details of these methods, there are differences in the effectiveness of image matching. In this paper, we select several state-of-the-art feature matching algorithms and compare them with classical matching methods to show the differences between them. We also show how the methods based on different principles perform in the face of different datasets and analyse the reasons behind these differences. Additionally, we evaluate the main methods in the context of image-based downstream vision tasks.

4.1 | Datasets

This section presents a collection of publicly available datasets that encompass diverse image transformations, allowing for a comprehensive evaluation of algorithm robustness and performance. The summarized information of the datasets used in the feature matching experiments is provided in Table 2, while Figure 14 illustrates example images from each dataset, except for YFCC100M.

(1) VGG [115]: This dataset comprises 40 pairs of images, including both planar scenes and scenes captured by a stationary camera. The dataset provides homography matrices for calculating corresponding coordinate relationships between image pairs.
(2) DAISY [116]: The DAISY dataset consists of wide-baseline image pairs along with depth images. It includes two short image sequences and several individual image pairs. A total of 47 image pairs are created for evaluation through intra-sequence image matching and pairwise image grouping.
(3) Remote sensing [79]: This dataset comprises 161 pairs of remote sensing images, including colour, infrared, SAR, and panchromatic photographs. It is commonly used for image-based localization, navigation, and change detection.
(4) YFCC100M [117]: The YFCC100M dataset is used in this study to evaluate the performance of different methods. It is a large-scale internet image collection that provides sparse reconstructions from structure-from-motion (SfM) and ground truth camera poses. It includes four scenes, with each scene containing 1000 images for evaluation.
(5) Aachen Day-Night [118]: The Aachen Day-Night dataset is employed to assess the visual localization accuracy of various methods. It consists of 4328 reference images and 922 query images (824 daytime images and 98 nighttime images), all captured in urban scenes.

TABLE 1 Comparison of some key matching algorithm mechanisms.

Methods	Time	Algorithm Mechanism	Advantage	Disadvantage	Applicable Scenarios
RANSAC [64]	1981	Resampling iteratively to obtain a minimal subset for establishing a parameterized geometric model to identify inliers	Computationally efficient and highly robust	With a high outlier rate, the computational efficiency experiences a significant decline	Short baseline stereo matching, wide baseline stereo matching, motion segmentation
ICF [77]	2010	Rejection of mis-matching by the correspondence function based on potential correspondence robust estimation	All types of images can be processed without the need to understand the deformation model	Accuracy decreases when complex transformations are involved	Image matching of rigid objects and non-rigid objects with unknown deformations
GS [72]	2010	Encode feature point information into graph edges and find correspondence based on the corresponding quadratic maxima	Robust to outliers, able to cope with one-to-one and one-to-many matching patterns	Scale factors cannot be estimated automatically and parameters need to be adjusted manually	Face matching, near-duplicate image retrieval
VFC [75]	2014	Solving for correspondence between two sets of feature points by interpolating vector fields	Maintains robustness for a large percentage of outliers (90%)	Recall and accuracy decrease when outlier rate increases	Rigid and non-rigid motion estimation
GMS [78]	2017	Transforming motion smoothness constraints into statistics to reject spurious matches	High matching speed with real-time	Weak recognition of rotational transformations of images	Real-time video matching
LFGC [89]	2018	The match is marked as an inside or outside point based on the multilayer perceptron acting on the pixel coordinates	Excellent testing standards can be acquired based on small training sets and limited supervision	Need to provide camera internal reference as input, and hypothesis generation based on a small subset will result in increased detection outliers	Wide baseline stereo vision matching
GFC [76]	2018	Using the Gaussian field criterion to remove false matches from potential matches	With linear time complexity	Complex transformations can lead to loss of accuracy	Remote sensing image alignment, image stitching
OANet [93]	2019	Building order-Aware network robust estimators to estimate inside point probabilities	Improved relative pose estimation accuracy on indoor and outdoor datasets	Different performance for different feature extraction methods	Relative pose estimation
LMR [94]	2019	Training classifiers to identify matches by using local neighbourhood structure consistency	Requires small training set and low time complexity	Small training set may result in less accurate matching results	Visual navigation, near-duplicate image retrieval
AdaLAM [85]	2020	Detection of outliers through a hierarchical process to achieve generalization with adaptive relaxation assumptions	Efficient use of modern parallel hardware for fast and accurate outlier filtering	Not applicable to widely repeated regular structures	Pose estimation, 3D reconstruction
ACNe [92]	2020	Adaptive clustering of hypothetical matches into several motion-consistent clusters as well as mismatched clusters	Achieving quasi-linear time complexity	High number of parameters required and high spatial complexity	Near-duplicate image detection, image segmentation
GLOF [84]	2021	Guided local anomaly factor identification and removal of mismatches based on multi-granularity neighbourhood structure preservation	Good performance for multiple viewing angles, light changes, shading, rotation, etc	Special training is required for complex transformations	Near-duplicate image retrieval
MAGSAC++ [69]	2021	Application of σ-consensus to generate estimation models by marginalizing a range of noise scales	Improved model quality by eliminating the need to manually determine sampling thresholds	Does not perform well at high outlier rates and is suitable for image matching with known transformations	Homography estimation, base matrix estimation
LSV-Anet [95]	2022	Proposed "local structure visualization attention (LSV)" network to convert outliers into dynamic visual similarity assessment	Consistent image matching accuracy performance with small training set	Global features of the match are ignored	Remote sensing image matching and photogrammetry

(Continues)

TABLE 1 (Continued)

Methods	Time	Algorithm Mechanism	Advantage	Disadvantage	Applicable Scenarios
LOGO [86]	2022	Construction of a locally oriented feature matching framework to identify inner points from the set of points to be matched	Robust for complex non-rigid transformations	Time-consuming when working with data sets containing a large number of false matches	Estimation of essential and basic matrices, image alignment
NASE [82]	2022	Mismatch elimination by imposing a kinematic consistency constraint to fit an affine subspace in the neighbourhood to approximate the interior point manifold	The optimization algorithm derives a log-linear time complexity and the multi-scale strategy allows the method to cope with more situations	Difficult to generalize to unknown scenarios as a data-driven approach	Base matrix estimation, real-time image matching
LGSC [96]	2022	Combining integer quadratic programming with compensating terms for resisted matching of local graph structure consistency to derive closed solutions with linear complexity for matching	Low time and space complexity	Relies on local topological representation; too high outlier rate can lead to computational failure; not very fast computation	Image Alignment
NMRC [81]	2022	Preserving the local neighbourhood structure between two feature points in a potential true match on a low-dimensional manifold	Linear time complexity to cope with complex image transformations	Cannot cope with areas of sparse potential correspondence	Remote Sensing Image Alignment
GLAM [107]	2023	Learning semantic features using intra-image attention and cross-attention for exact matching	Improved matching accuracy using graph pattern learning	Low generalization effect due to training on fixed objects	Image pattern mining

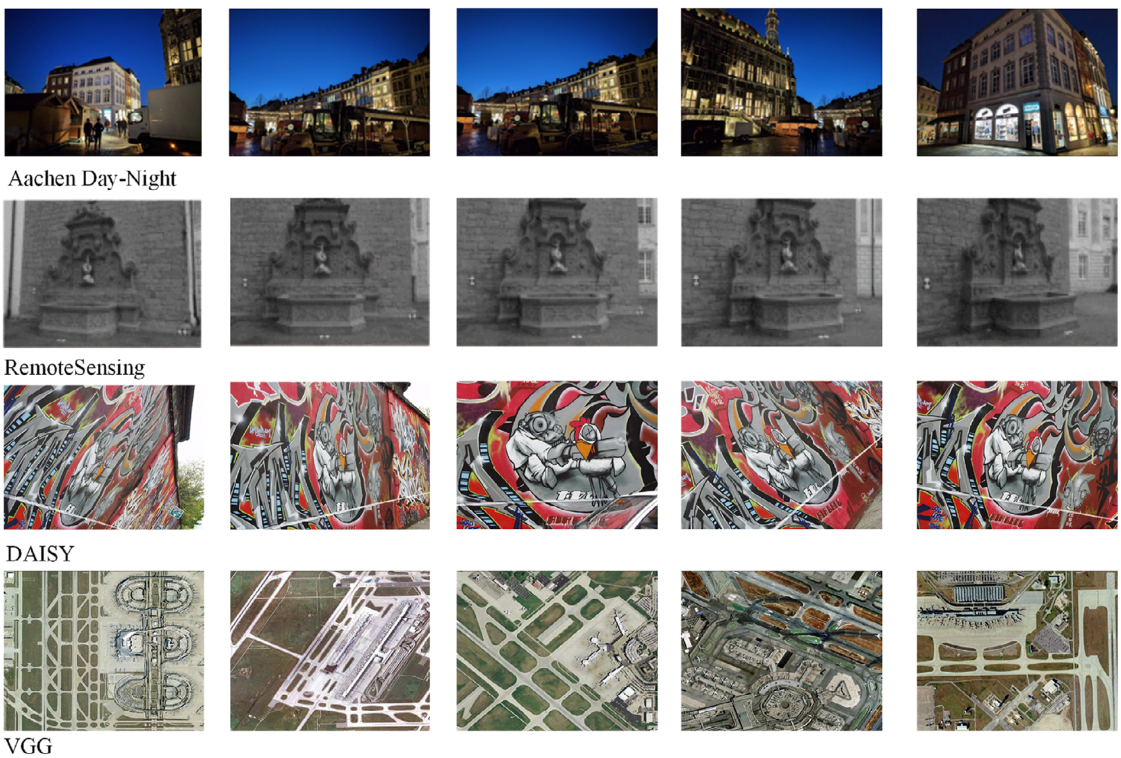


FIGURE 14 Example pictures of datasets.

TABLE 2 Datasets summary information.

Datasets	Number of image pairs	Groud Truth	Attribute
VGG	40	Homography matrix	Viewpoint and affine variations,rotation and blur
DAISY	47	Camera pose and depth	Wide baseline
RS	161	Camera pose	Image localization and change detection

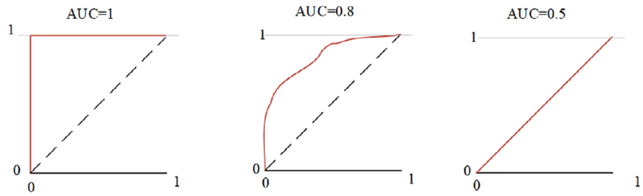


FIGURE 15 Corresponding curves for different areaS under the curves. AUC, area under the curve.

4.2 | Assessment criteria

The results of feature point matching are usually evaluated based on the number of correct and incorrect matches between image pairs, with precision (P) and recall (R) as quantitative metrics. The calculation formulas for precision and recall are as follows:

$$P = \frac{TP}{(TP + FP)} \tag{9}$$

$$R = \frac{TP}{(TP + FN)} \tag{10}$$

In this context, TP represents the number of True Positive samples, which are samples that are labeled as positive and predicted as positive. FP represents the number of False Positive samples, which are samples that are labeled as negative but predicted as positive. FN represents the number of False Negative samples, which are samples that are labeled as positive but predicted as negative.

Precision is the ratio of the number of correctly predicted matching pairs to the total number of predicted matching pairs in the matching results. Recall is the ratio of the number of correctly predicted matching pairs to the total number of actual matching pairs present.

In the pose estimation experiments, different thresholds (5°, 10° and 20°) of pose error area under the curve (AUC) are used to represent the error between the ground truth and the estimated results. AUC represents the area under the receiver operating characteristic (ROC) curve, where the ROC curve plots the true positive rate on the vertical axis and the false positive rate on the horizontal axis. By increasing the threshold for determining positive instances, a set of (x, y) points can be obtained and connected to form the ROC curve, with AUC representing the area under the curve in the lower right region. Please refer to Figure 15 for an illustration.

The figure displays the curves corresponding to three AUC values. AUC describes the probability of positive instances being ranked higher than negative instances and is used to describe the overall performance of a model, specifically in this paper, the relative pose prediction accuracy. Evaluation metrics in visual localization are similar to relative pose estimation, as they both represent the pose accuracy at different thresholds.

4.3 | Results on feature matching

This paper selects several state-of-the-art feature matching methods, including locally preserved matching (LPM) [79], deep learning-based local matching refinement (LMR) [94], learning feature-based geometric consistency (LFGC) [111], and the density-based GLOF method [84]. Additionally, classical hypothesis-verification-based RANSAC [64] and correspondence function-based ICF [76] algorithms are introduced for comparison. The deep learning-based methods are trained based on the original authors' implementation details, and the parameter settings are consistent with them. The generation of the assumed matching set is implemented using the SIFT method from the open-source VLFeat toolkit. Table 3 presents the average precision (P) and recall (R) results of these methods on the three datasets. During our experiments with certain methods, we used pre-trained models, which sometimes showed unexpected results on specific datasets. Additionally, some matching methods did not provide their source code, so we relied on the data reported in their papers. In Tables 3–5, any missing data due to these cases is denoted with "-."

4.3.1 | Results on DAISY

The DAISY dataset mainly consists of wide baseline images, and RANSAC, LPM, and LMR all achieve good precision and recall. Among them, RANSAC, as a classical outlier removal method, still exhibits high robustness for different types of samples but suffers from slow matching speed. Both the deep learning-based LMR and the traditional method of LPM achieve good matching results for wide baseline images by robustly extracting neighbourhood information from feature points. In the experiments, there are also cases of extremely poor matching performance for specific image pairs. From the table, although the precision of ICF is very high, the recall is very low at 37%. This may be due to the fact that the non-parametric model used by ICF is insufficient to discover all correct matches and can only detect matches that conform to specific patterns. GLOF exhibits the highest recall in the DAISY dataset, which is mainly because GLOF assigns a local outlier score to each corresponded pair and combines it with the preserved multi-scale neighbourhood structure for scoring and filtering, aiming to retain as many correct matches as possible.

TABLE 3 Performance of matching methods on different datasets. AUC: Area Under the Curve.

Method	Criteria	DAISY	RS	VGG
RANSAC [64]	P	95.97	97.20	87.75
	R	93.90	95.83	89.24
ICF [77]	P	94.71	88.93	92.35
	R	37.67	60.37	50.78
GS [72]	P	97.53	92.40	96.62
	R	72.16	86.35	95.31
VFC [75]	P	–	93.48	**98.14**
	R	–	95.69	95.87
GMS [78]	P	93.28	97.31	96.12
	R	86.61	88.53	82.27
GFC [76]	P	–	95.39	97.81
	R	–	96.96	97.57
OANet [93]	P	69.67	79.87	–
	R	95.64	99.02	–
LMR [94]	P	98.17	99.07	79.30
	R	95.19	99.14	81.22
LPM [79]	P	**98.29**	98.12	97.04
	R	96.10	98.81	85.79
AdaLAM [85]	P	98.11	**99.43**	–
	R	89.93	89.81	–
ACNe [92]	P	97.88	–	97.84
	R	86.06	–	91.00
LFGC [89]	P	97.63	98.12	94.86
	R	43.89	98.81	38.43
RFM-SCAN [83]	P	97.87	94.01	–
	R	96.20	94.16	–
GLOF [84]	P	95.36	98.39	90.24
	R	**97.86**	94.39	85.30
LSV-ANet [95]	P	–	97.20	–
	R	–	**99.56**	–
LOGO [86]	P	96.69	98.96	–
	R	93.95	95.82	–
NASE [82]	P	97.12	–	96.28
	R	97.31	–	**99.96**
LGSC [96]	P	93.80	97.58	–
	R	96.34	97.64	–
MAGSAC++ [69]	P	94.51	94.53	–
	R	90.91	99.34	–
NMRC [81]	P	97.23	99.39	–
	R	97.71	99.45	–

4.3.2 | Results on RS

LMR also exhibits high precision and recall on the RS dataset, confirming the effectiveness of the K-nearest neighbour strategy in capturing local neighbourhood structures. Moreover,

TABLE 4 Performance on YFCC100M.

Descriptor	Method	AUC		
		5°	10°	20°
SIFT [27]	NN [63]	15.19	24.72	35.30
	SGMNet [99]	35.63	55.40	71.95
	SuperGlue [52]	30.12	47.25	63.05
	ClusterGNN [97]	32.82	50.25	65.89
	PointCN [89]	47.98	58.13	68.67
	ACNe [92]	–	–	78.00
	LGLFM [111]	49.60	60.36	71.37
SuperPoint [58]	NN	14.12	28.86	44.90
	SGMNet	32.44	52.80	69.90
	SuperGlue*	39.02	59.51	75.72
	SuperGlue	34.93	55.31	72.34
	ClusterGNN	35.31	56.13	73.56
	PointCN	38.10	49.06	61.48
	AdaLAM [85]	40.20	49.03	59.11
	OANet [93]	48.80	59.06	70.02
ASLFeat [61]	NN	14.40	27.80	43.20
	SGMNet	32.22	52.53	70.16
	SuperGlue	37.50	58.30	75.07
	ClusterGNN	42.62	61.22	76.75
ContextDesc [53]	NN	57.90	68.47	78.35
	SGMNet	66.63	76.21	84.33
	SuperGlue	65.98	75.17	83.64
	PointCN	51.95	62.60	73.33
	AdaLAM	60.75	70.91	80.23
	OANet	62.28	72.56	81.80

LMR is suitable for dealing with different types of image datasets since the parametric geometric model is not used as a strict constraint. NMRC improves the matching performance in complex scenarios by designing an iterative filtering matching strategy based on neighbourhood similarity, which achieves satisfactory results. However, ICF still has the problem of low recall in RS datasets. AdaLAM exhibits high performance and optimal precision in the RS dataset, benefiting from its effective use of parallel hardware to obtain fast and efficient filters, making it suitable for scenarios requiring precise matching. As an end-to-end method, LSV-ANet shows strength in capturing as many correct correspondences as possible. This is achieved by continuously learning the matching process and adjusting the grid structure and scale for matching, leading to the derivation of the optimal grid regions for matching and discarding regions where correct matches cannot be found.

4.3.3 | Results on VGG

Both NASE and GFC methods achieve satisfactory results. NASE improves accuracy by introducing a density-based seed

TABLE 5 Performance on Aachen day–night.

Descriptor	Method	0.25m, 2°	0.5m, 5°	5m, 10°
RootSIFT [43]	MNN	43.9	56.1	65.3
	OANet [93]	69.4	83.7	94.9
	SuperGlue [52]	63.3	80.6	98.0
	SGMNet [99]	70.4	85.7	98.0
ContextDsec	MNN	65.3	80.6	90.8
	OANet	74.5	86.7	99.0
	SuperGlue	77.6	86.7	99.0
	SGMNet	75.5	87.8	99.0
	PointCN	75.5	85.7	98.0
SuperPoint	MNN	71.4	78.6	87.8
	OANet	77.6	86.7	98.0
	SuperGlue*	79.6	90.8	100.0
	SuperGlue	76.5	88.8	99.0
	SGMNet	77.6	88.8	99.0
	PointCN	75.5	89.8	99.0
	S2DNet	74.5	84.7	
End2End Methods				
Patch2pix [55]		79.6	87.8	100.0
D2-Net [59] + Point–CN [89]		76.5	87.8	99.0
DualRC-Net [110]		79.6	88.8	–
SparseNCNet [109]		76.5	84.7	–

selection strategy that enhances features and increases the likelihood of finding the correct match. GFC enhances robustness to different transformations by employing two different approaches for rigid and non-rigid transformations and applying them to various transformations in the VGG dataset. VFC demonstrates high recall rates in the VGG dataset, which may be attributed to its initial reduction of the potential matching set size using a non-parametric model in the outlier removal process, followed by the estimation of geometric parameters for further outlier removal. However, VFC performs poorly in the wide-baseline DAISY dataset, which may be attributed to the large variations in image angles that make it difficult to form a smooth vector field. On the other hand, although LFGC exhibits good precision and recall in the RS dataset, it does not perform well in the VGG dataset. Despite its high precision, the recall rate is only 38% because LFGC's primary objective is to identify well-matched correspondences for more accurate construction of the transformation matrix between two point sets, which results in the removal of some correct matches that do not meet the objective.

4.4 | Results on pose estimation

The purpose of camera pose estimation is to determine the orientation of the camera in an object or scene. The task of indoor image matching is relatively challenging due to a lack of texture, high self-similarity, complex 3D scenes, and large changes in viewpoints. Compared to traditional methods, deep learning-based feature matching methods perform better and have more applications in this field. In this study, we selected several state-of-the-art neural network methods, such as SuperGlue [52] and SGMNet [99], and compare them with various classical feature descriptors. Among them, SuperGlue* is the matching model trained using the SuperPoint descriptor by the original author.

In the experiment, the pose error AUC at different thresholds (5°, 10° and 20°) was obtained, which measures the maximum angular difference between the estimated results and the ground truth values in terms of rotation and translation. For the provided matches from different methods, the essential matrix was estimated using RANSAC to compute the relative pose.

As shown in Table 4, under the scenario of SIFT-based features, LGLFM outperforms other methods. As a guided matching approach, LGLFM can align the maximum number of images when guiding SIFT features, resulting in higher feature point accuracy compared to other methods. ACNe, as a method belonging to attention-based contextual networks, also demonstrates outstanding performance in indoor matching. However, RANSAC can somewhat compromise its performance. If combined with MAGSAC, ACNe exhibits even greater superiority compared to other methods. After applying RANSAC, SuperPoint features are more suitable for OANet, with performance similar to that of SuperGlue. OANet introduces DiffPool and Order-Aware DiffUnpool layers to cluster meaningful nodes and capture local context. Furthermore, OANet's order-aware filtering block can capture global context, thereby improving matching accuracy for indoor and outdoor datasets. ClusterGNN outperforms other methods using ASLFeat, indicating that this matching approach is more suitable for learning-based feature descriptors. It also performs well when using other features, demonstrating ClusterGNN's good approximation and denoising capabilities. SGMNet performs optimally in ContextDesc feature points and remains stable across other feature points. As a deep learning method, SGMNet offers higher accuracy with moderate computational and memory costs, making it suitable for matching tasks with a large number of key points.

4.5 | Results on visual localization

Visual localization is an important application of feature matching. To evaluate the performance of various matching methods in outdoor localization, this study utilizes the Aachen Day-Night dataset. Table 5 reports the pose estimation accuracy at different thresholds, considering factors such as changes in scene angles and day–night variations. From the table, it can be observed that SGMNet outperforms SuperGlue and OANet in most metrics when using ROOTSIFT and ContextDesc descriptors. SGMNet employs a seed graph matching network with a sparse structure, which reduces redundant connectivity in the matching process, significantly lowers computational complexity, and enhances matching accuracy. SuperGlue still demonstrates its advantage when using SuperPoint feature points, indicating its compatibility with visual localization tasks. ACNe and ClusterGNN also exhibit commendable performance in visual localization experiments. Both networks leverage self-attention and cross-attention mechanisms to construct attention-based architectures for learning the matching task, highlighting the critical role of such mechanisms in finding correct matches. Additionally, DualRC-Net, as an end-to-end dual-resolution architecture network, establishes dense pixel correspondences in a pair of images. It utilizes the established 4D interrelationship to guide the model for robust matching, enhancing both the reliability and localization accuracy of the matches.

5 | CHALLENGES AND FUTURE TRENDS

5.1 | Challenges

In recent years, feature matching has evolved from simple traditional manual matching to combining with deep learning or even relying entirely on neural networks. However, as an ongoing research topic, feature matching still faces a number of challenges that require further improvement and refinement.

(1) Feature extraction is uncontrollable: Feature-based matching methods exhibit good stability, but how to detect and describe features efficiently, as well as how to perform feature matching and geometric model estimation, are challenging issues. The extraction algorithms are unable to controllably extract feature points with the same 3D location in the real world between images, thus ensuring the repeatability of the feature point set and the matchability of the description set. Moreover, due to some omissions and errors in the feature detection process, current mismatch removal methods are still insufficient for dealing with multi-view, wide baseline and deformable motion scenarios.
(2) Descriptors are not entirely reliable: During the matching process, matching N detected feature points in the source image with N detected feature points in the target image results in $N!$ possible match pairs, leading to high computational complexity and including outliers and noise. From traditional, handcrafted extraction methods like SIFT [27] to deep learning methods like SuperPoint [58], various local descriptors have been continuously proposed. However, relying solely on descriptors (the local appearance information of feature points) inevitably leads to incorrect matches.
(3) Difficulties remain in matching occluded and locally varying scenes: Occlusion and local variations are common issues in feature matching, which seriously affect feature point extraction and subsequent matching. Most of the current methods can solve the global transformation and simple local variation problems, but they are incompetent in dealing with occlusion and complex local variation.

5.2 | Future trends

(1) Incorporating semantic information for feature extraction: Semantic information involves utilizing object category attributes and other relevant information in the image. In the future, it is possible to combine semantic information with image features and use neural networks to learn and extract features, thus enabling controlled extraction of desired features.
(2) Preprocessing of images before matching: By employing image segmentation techniques, useful regions can be extracted based on different image categories, thus avoiding noise and irrelevant regions. This approach limits the matching process to descriptors within the identified useful regions, thus alleviating the problem of high computational complexity caused by a large number of potential matches. To effectively extract the useful regions, the integration of neural networks into learning and decision-making can be explored.
(3) Utilizing cross-attention to aggregate contextual information on a graph: Attention mechanisms have been widely employed in visual tasks, and combining contextual attention can generate multiple paths that emphasize both locally and globally important contextual information. By combining pooling strategies and circular convolutions, reliable graph contexts can be obtained, allowing for effective inference of inliers even in scenarios with significant local variations.
(4) Integration of local and global methods: Among mismatch removal methods, local feature-based approaches encounter difficulties when handling repetitive textures and high outlier rates, while global feature-based methods perform poorly when confronted with localised distortions and individual motions, which creates a need for methods capable of handling large-scale scenes. These problems can be alleviated by combining local and global geometric consistency. However, the process of scaling from local to global introduces high computational complexity, and therefore algorithms are needed to optimise the computational steps.

6 | CONCLUSION

Feature matching is a crucial strategy in image processing with numerous downstream applications. We provide an overview of the development of feature detection and description, with a focus on feature matching methods, including both traditional methods approaches and learning-based methods. By analyzing the principles of matching algorithms, we discuss their advantages, and disadvantages, and recommended application scenarios. Combining evaluation metrics and application areas of feature matching, we select advanced and classical methods for experimental comparison and analyse the results.

Deep learning based matching methods are currently evolving, and some of them have achieved better performance than manual methods. Additionally, feature detection and description methods based on neural networks are becoming more mature, supporting matching networks to achieve better performance. However, when facing the challenges of matching scenarios such as complex image transformations, more robust and efficient methods are still needed to achieve better matching results.

AUTHOR CONTRIBUTIONS

Qian Huang: Writing—original draft. **Xiaotong Guo**: Methodology. **Yiming Wang**: Data curation. **Huashan Sun**: Software. **Lijie Yang**: Methodology.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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